eCommerce Database Structure, Marketplace Readiness & SKU Logic

1. Database Tables & Relationships

Departments
- department_id (PK)

- name

- description

- status

Categories
- category_id (PK)

- department_id (FK)

- name

- description

- status

SubCategories
- subcategory_id (PK)

- category_id (FK)

- name

- description

Products
- product_id (PK)

- subcategory_id (FK)

- product_name

- brand

- description

- base_price

- status

ProductVariants

- variant_id (PK)

- product_id (FK)

- sku

- colour

- size

- model

- material

- price

- stock_qty

ProductImages

- image_id (PK)

- variant_id (FK)

- image_url

- is_primary

Attributes

- attribute_id (PK)

- name (Colour, Size, Model, Material, etc.)

VariantAttributes

- variant_attribute_id (PK)

- variant_id (FK)

- attribute_id (FK)

- value

Filters

- filter_id (PK)

- category_id (FK)

- attribute_id (FK)

Inventory

- inventory_id (PK)

- variant_id (FK)

- stock_qty

- warehouse_location

2. Table Relationships (ERD Logic)

- One Department → Many Categories

- One Category → Many SubCategories

- One SubCategory → Many Products

- One Product → Many Variants (SKU level)

- One Variant → Many Images

- One Variant → Many Attributes

- One Category → Many Filters

3. SKU & Variant Logic

- SKU is generated per variant, not per product

- SKU structure example: CAT-BRAND-MODEL-COLOUR-SIZE

- Example SKU: SHOE-NIKE-AIRMAX-BLACK-42

- Each colour/size/model combination is a unique SKU

- Stock is managed at SKU (variant) level

4. Filter Logic (Search & Navigation)

- Filters are category-specific

- Common filters: Price, Brand, Colour, Size

- Dynamic filters pulled from VariantAttributes

- Only show filters that exist in category data

- Supports Takealot, Shopify & WooCommerce filtering models

5. Marketplace Readiness (Takealot / Shopify / WooCommerce)

- Each Product = Parent product

- Each Variant = SKU / Child product

- Attributes mapped to marketplace attributes

- Images mapped per variant
- Inventory synced at SKU level
- Supports CSV bulk uploads
- Supports API integration